

TOWARDS A NETWORK THEORY OF THE IMMUNE SYSTEM

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The essence of society is repression of the individual.

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(Life against Death)

I. — INTRODUCTION

While listening to this week's colloquium on *the genetics of immunoglobulins and of the immune response*, I have given some thought to the long road along which immunology has travelled since the days of Louis Pasteur. I should like to place my concluding remarks in this perspective, by briefly looking at the past, and at our present situation, and by attempting to predict in what direction immunological theory will develop in the next years.

The development of a science depends on the thoughts and interactions of successive generations of scientists; by their work, these scientists uncover additional phenomena and facts, expanding the territory and the number of problems to be considered. A steady rate of progression of a science therefore requires something like an exponential growth in the number of people engaged. This is what has happened in immunology over the past hundred years. I shall illustrate the growth with some numbers that are rough estimates, and starting with the year 1970, I shall go back in time by steps of twenty years.

The number of research workers in immunology in 1970 was estimated to be about 10,000. Assuming that the number of immunologists in the world has, on the average, tripled every twenty years, there were 3,300 immunologists in 1950, 1,100 in 1930, 360 in 1910, 120 in 1890, and 40 in 1870. I now quite arbitrarily introduce the notion of « outstanding immunologist » and estimate that, on the average, among 40 immunologists there will be one that is « outstanding ». This would mean that at present, or in 1970, there are 250 outstanding immunologists. This number makes sense, I think. In fact, many of these are now present in this

room. Going back in time, we would expect to find about 80 outstanding immunologists in 1950, 27 in 1930, 9 in 1910, 3 in 1890 and 1 in 1870. This one, obviously, was Louis Pasteur, the father of immunology, to whom the present colloquium is dedicated.

II. — PROGRESS OF IDEAS

I shall now make an equally brief sketch (summarized in table I) of the way I see the development of ideas in immunology during these successive periods of 20 years, and shall start out with the period 1870-1890 which is the time during which Pasteur has turned his attention to immunology.

TABLE I. — Progress of ideas.

Aims	Periods	Pioneers Types of theory	Notions
Application	1870-1890	Pasteur Metchnikoff	Immunization Phagocytosis
Description	1890-1910	Behring Ehrlich	Antibodies Cell receptors
	1910-1930	Bordet Landsteiner	Specificity Haptens
Mechanisms	1930-1950	Sub-cellular	Antibody synthesis Antigen template
	1950-1970	Cellular	Selection Clones
System analysis	1970-1990	Multi-cellular	Network Cooperation and suppression

Practical applications of a science often precede the birth of this science itself. Gun-powder and the compass were invented and used long before chemistry and physics developed into sciences. Before Pasteur, knowledge of immunology was restricted to smallpox vaccination. The efforts of Pasteur were likewise directed towards immediate application of the phenomenon of immunity in preventive medicine. His great contribution was to demonstrate that Jennerian vaccination was only a special case that could be generalized to apply to many other microbial infections. A brilliant generalization such as this was needed and sufficient to create the science of immunology.

The period of Pasteur was followed by about forty years which I would briefly call the early descriptive period. During this time a large range of immunological phenomena were described, outlining, as it were, the territory with which this science would have to deal. Elie Metchnikoff

discovered phagocytosis and emphasized cellular aspects, whereas the existence of antibodies in serum was discovered by Behring and Kitasato in 1890, and applied in therapeutic medicine that same year. A dominating figure in the period 1890-1910 was Paul Ehrlich who extended the phenomenon of specific immune responsiveness far beyond the special cases of responses to microbial antigens; he coined the term « horror autotoxicus » to describe the astonishing fact that we do not seem to make antibody to our own tissue components; and he originated the field of tumour immunology. Even the insight that transplantation problems belong to immunology originated in that period (Schöne, 1912).

There was a number of outstanding immunologists in the period 1910-1930 among whom Jules Bordet was one of the greatest. This period brought a further expansion and characterization of the notion of immunological specificity, particularly through the work of Landsteiner. Not only did antibody specificity turn out to be exquisite, but the immune system was shown capable of producing specific antibodies against artificially synthesized chemicals that had never existed in the world.

Though application and description of immunology of course continued, the time in 1930 was ripe for the next period of 40 years during which attention was turned towards discovering the mechanisms underlying the immune response, particularly antibody formation. The theoretical proposals originating during the first 20 years of that period, 1930-1950, were mainly of a sub-cellular type. They focussed on the biosynthesis of antibody molecules. It was clear that antibody molecules were made by cells. It was equally clear that a producing cell could have no prior information regarding the antigen, most obviously so for the case of artificial antigens. The conclusion seemed to be inescapable, that the antigen must bring into the cell information concerning the complementary structure of the antibody molecule.

The following 20 years, 1950-1970, saw the decline of these early antigen-template theories of antibody formation, in favour of selective theories. We might now say that the most obvious reason for this transition lies in the fact that antibody molecules are synthesized like other proteins, and that the template for their amino acid sequence is not antigen but messenger-RNA. Historically, of course, the selective theories emerged before this was known, and before antigen-template mechanisms were proven wrong. The prototype of these theories is, of course, the clonal selection theory of Burnet [4] which is a cellular theory. We are still standing, at least with one leg, in this theoretical period and we try to interpret experimental findings, including those presented at this colloquium, within this theoretical framework. I shall therefore recapitulate some essential points of this cellular hypothesis, in order to evaluate whether it continues to deal satisfactorily with present-day experimental findings.

This will then lead me to a prediction concerning the theories that will be developed in the period 1970-1990 which, I think, will be of a multi-cellular type of which attempts to deal with the interaction of T cells and B cells represent early beginnings. Though the search for mechanisms at the

sub-cellular, cellular and inter-cellular levels will of course continue, I think that emphasis will shift to a structural analysis of the entire immune system.

III. — CLONAL SELECTION

The key postulate of the theoretical framework established in the period 1950-1970 is one of the very few general laws pertaining to the immune system. It is the law that one cell makes only one antibody. This law can be formulated with greater precision. For example, by saying that the immune system contains a population of antigen-sensitive lymphocytes each of which is already committed to the synthesis of antibody molecules of a single specificity and, upon stimulation, will transmit to all the progeny cells of its clone that same single commitment. It can be argued, of course, that the switch from IgM to IgG production within one clone of cells, and the findings reported by Herzenberg [1], that cells presenting IgM receptors can become IgG or IgA producers, and by Pernis and Rowe, that one lymphocyte can simultaneously present IgM and IgD receptors, all indicate that the law one cell $\rightarrow$ one antibody can refer only to the variable regions of the antibody molecules, or to the expression of v-genes. Furthermore, individual cells of one clone can change their inherited commitment by mutation, for which Milstein [1] brought experimental evidence. Attempts can be made to define the stage differentiation of a cell at which its single commitment becomes fixed. There must obviously be non-committed stem cells which are precursors of antigen-sensitive cells. In spite of such refinements, the law remains generally valid that all antibody molecules synthesized by a single cell and its progeny cells have identical combining sites because all cells of a clone express the same pair of v-genes for heavy and light chains. This law is now supported by many experimental findings, such as the failure of attempts to show that one cell produces antibody molecules of more than one specificity, and the success of removing responsiveness to a given antigen from a suspension of lymphocytes, by removal of a small fraction of precommitted cells that recognize this antigen. It should be noted that the latter finding was found to apply to T cells as well as to B cells. Herzenberg pointed out [1] that the positive demonstration has also been made, namely to retrieve responsiveness to the antigen in the small fraction of cells that were specifically removed.

As Pernis showed, the law one cell $\rightarrow$ one antibody leads to several predictions.

A) Allelic exclusion. This was said by Todd [1] to be one of the major contributions of allotype research.

B) All antibody-like receptors displayed by a lymphocyte should be identical or at least have identical light chains and identical heavy chain variable regions. Studies by fluorescent labelling of B cell receptors support this implication.

C) All antibodies produced by a single cell and its progeny should have the identical idio-*type*. The experiments on suppression of particular idio-*types* reported at this colloquium by Nisonoff [8, 11, 12] support this implication.

D) Among the lymphocytes of one animal the diversity with respect to precommitment must be equal to the diversity of the antibodies this animal can produce. In other words, the repertoire of antibody specificities and idio-*types* available to an animal equals the repertoire of lymphocytes expressing different pairs of v-genes.

This last point will now be conceded by almost everybody, I believe. During the last years of the period 1950-1970 this led to a controversy among immunologists concerning the origin of the v-genes expressed by lymphocytes. The proponents of the germ-line theory briefly argue as follows: all genes present in any somatic cell are copies of genes that were present in the zygote; *ergo*, the v-genes expressed by any lymphocyte are germ-line genes. The proponents of the somatic theories answer: this is right, except that genes can be modified in somatic cells, for example by mutation; they are staggered by the size of the immunological repertoire and do not believe that the number of v-genes required for encoding this repertoire could arise, nor be maintained, in evolution; they, therefore, propose that the zygote contains a small set of v-genes, and that the large set of v-genes expressed by the population of lymphocytes has arisen from the selection of somatic mutant cells. Both Milstein's and Oudin's reports at this colloquium lend some slender experimental support to these ideas [1].

This controversy dominated theoretical immunology during the last years of the 1960's, and all arguments *pro et contra* have been exhaustively reported in the literature [7]. I shall therefore turn my attention to other matters, except for noting that the clonal selection theory, or the law one cell $\rightarrow$ one antibody, is compatible with any view of the origin of the v-genes expressed in the lymphocyte population.

IV. — RECENT ADVANCES

Looking back at the papers presented at this colloquium, it strikes me that in 1960 practically nothing was known about the subjects discussed, and that in 1950 most of these subjects did not even exist.

Nothing was known of the primary structure of antibody molecules nor of allotypes and idio-*types*; the genetic determination of specific immune responsiveness and recognition had hardly begun to be explored; intracellular antibody synthesis was misunderstood. The account that Askonas presented [1] shows the enormous advances in knowledge of antibody biosynthesis achieved in the past twenty years. What now seems most incredible is that it was hardly even suspected twenty years ago that the small lymphocytes had anything to do with the immune system, whereas we now know that they *are* the immune system, or at least 98 % of it. I still

count it as a point in my own favour that I suggested this in 1955, and likewise prematurely proposed antibody diversity to be generated in the cells of the thymus [16]. Experimental proof of the involvement of lymphocytes and thymocytes came only in the 1960's through the work of Gowans, Miller, Nossal, and many others, so that we can now state quite simply that the immune system in an adult person consists of about 10^{12} lymphocytes and about 10^{20} antibody molecules. It is clear that a science that advances as rapidly as immunology requires a continuous revision of its theoretical foundations as well as a continuous expansion of the theoretical framework in order to encompass phenomena that older theories did not have to deal with.

1) Let us start with the primary structure of antibody polypeptide chains. Last year's Nobel Prizes testified to the enormous importance of this structural work to all immunological thinking. I remember the delight with which in 1963 we learned of Hilschmann's and Craig's demonstration of the sequence variability among the 107 N-terminal amino acid residues of two light chains. In one stroke, a structural basis had been provided for a potentially enormous variation in antibody specificity. Porter and Milstein [1] have pointed out the usefulness of further sequence work for establishing genetic markers that can serve in determining allelic polymorphisms in the structural c- and v-genes, for proving that two genes code for one polypeptide chain, and for clarifying the genetic origin of the multiple variable regions. These studies are pertinent to the theories of phylogenetic versus ontogenic generation of antibody diversity that we have already briefly mentioned.

2) Serological determinations of allotypes and idiotypes have proven to be useful and illuminating experimental alternatives in the common situations where antibody heterogeneity prohibits the acquisition of sequence data. As Todd pointed out [1] allotype research has established genetic linkages between the structural v- and c-genes for antibodies as well as the allotypic identity of the receptor molecules on a single B cell and of the antibody molecules produced by one cell, thus lending additional support to the law one cell $\rightarrow$ one antibody. Idiotypes which are determined by the same variable regions that determine antibody combining sites are, in my opinion, of particular importance for understanding immunology, and I consider Oudin's report of the recent work of himself and his colleagues [1] among the most fascinating developments of these years. Mage, Herzenberg and Nisonoff briefly discussed allotype suppression and idiotypic suppression, and I intend to return to these essential phenomena in the following sections, where I shall also attempt to reconcile with current immunological concepts the apparently paradoxal findings reported by Oudin [1] that antibodies directed against different determinants on the same antigenic molecule are apt to have similar idiotypes and that immunization results in the parallel production of unspecific immunoglobulin molecules of the same idiotypes as the specific antibodies [5, 6, 26].

3) Another apparent paradox that came to light at the end of the

1950-1970 period was the finding of a genetic determination and simple Mendelian inheritance of specific immune responsiveness. The experimental facts were presented by Biozzi, Benacerraf and McDevitt in this colloquium. So-called Ir-genes that are closely linked to the histocompatibility locus exert a dominant control of the response of T cells to specific antigens, and, through the failure of T cell cooperation, influence the antibody response by B cells to these antigens and to haptens coupled to them [1]. This seems compatible neither with a random somatic generation of antibody diversity, nor with a germ-line theory of antibody diversity, as the Ir-genes are not linked to the structural genes for antibodies. In discussing the implications of these findings, Benacerraf and McDevitt regarded inescapable the conclusion that there exists a class of molecules encoded by these Ir-genes, which are responsible for the recognition of specificity at the T cell level, and that these molecules are not immunoglobulins. A serious difficulty to be dealt with is the large repertoire of T cell specificities which would seem to require a larger number of structural genes than the relatively limited region of the genome, the major histocompatibility complex, can harbour. Other structural genes for lymphocyte surface proteins located in this genome region are the SD-genes (for serologically determined histocompatibility antigens) and LD-genes (for the antigens that determine T cell responses in mixed lymphocyte cultures). These two sets of genes are dominant in determining tolerance, *i. e.* the possession of one allele leads to tolerance of lymphocytes to the product of this allele. We could imagine that the presence of an Ir-product on the T cell surface likewise engenders tolerance towards this product. All that would then be needed is to postulate that the acquisition of tolerance to an Ir-product leads to T cell responsiveness to a set of antigens [18]. The expression of this need, it may be said, is not a scientific theory. But then neither is the expression of the need for an unidentified class of non-immunoglobulin recognition molecules. Benacerraf and McDevitt called their proposal revolutionary and heretical, respectively, and I suggest that this proposal should be resisted as long as other ways of interpretation remain possible, that do not sacrifice the idea that immunological specificity resides solely in antibody molecules.

4) Striking advances have been made in the last few years in characterizing sub-populations of lymphocytes. Gowans, Miller and Nossal presented the latest findings in this population analysis [1]. Nossal produced further evidence for the notion that the law one cell $\rightarrow$ one antibody applies equally to T- and B- cells, and for the belief that the functional receptor on T lymphocytes is an IgM-like molecule. This view is supported by accumulating experimental evidence [29]. Furthermore, Gowans [1] showed that not all functional B cells have easily demonstrable immunoglobulin receptors, whereas McDevitt [1] found that anti-IgM inhibits antigen binding by T cells. Moreover, IgM bearing T cell lymphomas are known to exist. Nossal and Benacerraf seemed willing to allow for the possibility that T cells could make use of two different recognition systems, one based on immunoglobulins and the other on the Ir-gene products. This kind of compromise, I believe, will not remain tenable. Miller claimed a key role

for the T cell in the immune response and almost relegated the B cell to a class of secondary effector cells. To me, the most striking facts remain the dichotomy of lymphocytes into T cells and B cells, and the demonstration of cooperative as well as repressive interactions between cells of these two populations.

Though the T cell/B cell dichotomy does not directly contradict the theoretical foundations that were established during the period 1950-1970, it certainly shows that this theoretical frame needs widening in order to remain a useful basis for the interpretation of experimental findings in immunology.

V. — FORMAL NETWORK

I shall now attempt to predict the type of theoretical developments that I foresee for the period 1970-1990. By this, I do not mean to set forth a set of partial hypotheses to explain the signal or the constellation of signals to which an antigen-sensitive lymphocyte either responds or does not respond, nor to explain mechanisms of interaction among and between T cells and B cells, or of interactions between antibody molecules and cells, nor to deal with the effector mechanisms that may become operative as the results of such interactions.

The determination of mechanisms will remain important, but I think there is now a need for a novel and fundamental idea that may give a new look to immunological theory, similar to the impact that the idea of selection had on theoretical developments in the period 1950-1970. Before coming to this, I shall make a few suggestions on terminology and then start out by considering: (I) Repertoires, (II) Dualisms, and (III) Suppression.

Immunological terminology is sometimes clumsy and I now take up a proposal I made many years ago [8], namely to replace the term « antigenic determinant » by « *epitope* », and the term « antibody combining site » by « *paratope* ». An epitope is a patch on an antigen molecule, which presents a certain relief or pattern that can be recognized with various degrees of precision by complementary patterns of paratopes located on antibody molecules. Epitopes and paratopes are two essential features of the immune system enabling it to accomplish its task which is pattern recognition. Two outstanding accomplishments of the recent period are the discoveries of allotypes and idiotypes, or the finding that antibody molecules present epitopes and that these can play a functional role, as experimentally demonstrated by allotype and idiotypic suppression. I draw particular attention to idiotypes which were defined by Oudin as the antigenic specificities of antibodies produced by an individual or a group of individuals in response to a given antigen [1]. An idiotypic thus is a set of epitopes displayed by the variable regions of a set of antibody molecules. Each single idiotypic epitope I shall call « *idiotope* ». An idiotypic then denotes a certain set of idiotopes. The patterns of idiotopes are determined by the same variable

regions of antibody polypeptide chains that also determine the paratopes. In short, each Fab arm of an antibody molecule displays one paratope and a small set of idiotopes.

I a) Let us now come to the « *repertoires* », and first consider the repertoire of antibody combining sites or the total number of different paratopes in the immune system. We can here distinguish between the total number of paratopes that could potentially arise in an individual possessing a given genome, and the number of paratopes that are actually present in a given immune system [19]. The « potential » repertoire is obviously far larger than the « available » repertoire which is the number of different paratopes in the immune system of an individual at a given moment. I have previously tried to estimate the size of the available repertoire [16, 17], which may obviously vary in different species of animals and between different individuals. For the moment I shall simply assume that the available repertoire of paratopes in a given immune system is of the order of several millions.

I b) In considering the repertoire of idiotopes, I refer to the experiments of Oudin *et al.* [27, 28] and those of Kelus and Gell [20], in which the idiotypes of antibodies produced by different rabbits to the same bacterial vaccine were studied. These studies, as well as those by Eisen, Kunkel and others on the idiotypes of murine and human myeloma proteins [22, 23] suggest that the repertoire of idiotopes is enormously large. This conclusion is supported by the findings reported by Nisonoff [1] that a given idiootype is exceedingly rare in normal serum globulin. For the moment, I shall simply assume that the repertoires of paratopes and of idiotopes are both of the same order of magnitude. The studies of Askonas, Williamson, Wright and Kreth [3] on the differences in iso-electric point between antibody molecules produced by mice of one inbred strain to the same antigen clearly suggest that the potential repertoires of this mouse strain are far larger than the repertoires that are apparently made use of by a single mouse of this strain. The demonstration that animals can be stimulated to make antibodies to the idiotopes of antibody molecules produced by other animals of the same species or strain, indicates that a given animal possesses available paratopes that can recognize any idioptope occurring in the species. I therefore find it reasonable to assume that, within the immune system of one given individual, any idioptope can be recognized by a set of paratopes and that any paratope can recognize a set of idiotopes. I have tried to show in an earlier paper [19] that this recognition of antibody molecules by other antibody molecules in the same serum will result in only a negligible concentration of complexes if the product of the concentrations of reacting paratopes and idiotopes remains sufficiently small. This proviso can be satisfied by assuming sufficiently large repertoires of paratopes and idiotopes. If we accept these propositions we can conclude that, formally, the immune system is an enormous and complex network of paratopes that recognize sets of idiotopes, and of idiotopes that are recognized by sets of paratopes. I here make the additional assumption that the presence of

identical paratopes on two antibody molecules does not necessarily imply that these molecules present identical idiotopes.

II) The second point I need to make is that the immune system displays a number of *dualisms*. The first of these is the occurrence of T lymphocytes and B lymphocytes with partly synergic, partly antagonistic interactions. The second dualism is the one we have just discussed, namely that antibody molecules can recognize as well as be recognized. These properties lead to the establishment of a network, and as antibody molecules occur both free and as receptor molecules on lymphocytes, this network intertwines cells and molecules. The third dualism I draw attention to is the ability of an antigen-sensitive lymphocyte to respond either positively or negatively to a recognition signal. A positive response leads to cell proliferation, cell activation and to antibody secretion, a negative response results in tolerance and suppression. We do not know the precise conditions that determine whether the recognition of an epitope by the paratope of a cell receptor will lead to a positive or a negative response. It may be that the presence of both IgM and IgD receptors on single B cells (described by Pernis and Rowe) is related to this potential dualism of response. For certain foreign epitopes the induction of high zone and low zone tolerance and of stimulation in an intermediate zone have been demonstrated. For the recognition of idiotopes by paratopes we have the findings of Iverson and Dresser [14] that the injection of plain myeloma protein into a mouse leads to tolerance to the idio-type of this immunoglobulin, whereas polymerized or modified myeloma protein can induce an anti-idiotypic antibody response. Thus, a lymphocyte can respond in two ways when the paratopes of its receptor molecules recognize an epitope, or an idio-type.

What happens to a lymphocyte in the reverse situation when the idiotopes of its receptor molecules are recognized by the paratopes of free antibody molecules or by the paratopes of the receptors of another cell? Most experimental findings suggest that the lymphocyte is then repressed.

III) My third point is to stress the importance of *suppression*. I have become increasingly convinced that the essence of the immune system is the repression of its lymphocytes. Lymphocytes are suppressed by anti-allotypic antibody [24]. As Herzenberg and Jacobson have shown [15], this allotype suppression is maintained by T cells that recognize the allotype of B cell receptors. The situation is similar for the suppressive effect of anti-idiotypic antibody and, by analogy, T cells recognizing the idiotopes of B cell receptors may be assumed likewise to maintain B cell suppression. Conversely, we could conclude that, normally, B cells remain functional because of the absence of sufficient numbers of specific suppressive T cells, or in other words, that B cells can suppress the emergence of recognizing T cells in certain situations. In general, this seems to lead to some dynamic equilibrium or steady state among the elements of the immune system. As tolerance induction shows that B cells and T cells can also be suppressed

when the paratopes of their receptors recognize a foreign epitope or an idiotope, the number of negative, suppressive situations appears to exceed the number of positive, stimulatory situations.

VI. — FUNCTIONAL NETWORK

Even if conceding the possibility of viewing the immune system as a formal network, one may question whether the concentrations of individual paratopes and idiotopes are sufficient to establish an internal functional network. Experiments have shown that the concentrations of epitopes needed to induce low-zone tolerance are often far lower than what most of us would estimate to be the average concentration of a single paratope or idiotope in the system. If, for example, we assume a repertoire size of five millions, both for paratopes and for idiotopes, then the average concentration in our blood of antibody molecules representing one member of a repertoire would be 10^{10} per ml, whereas low-zone tolerance has been achieved with epitope concentrations ranging from 10^6 - 10^{12} per ml [2, 25, 31].

We conclude that the immune system, even in the absence of antigens that do not belong to the system, must display an eigen-behaviour mainly resulting from paratope-idiotope interaction within the system. By its eigen-behaviour the immune system achieves a dynamic steady state as its elements interact between themselves, and as some elements decay and new ones emerge. A complete network theory of the immune system would require greater precision in stating the expected results of the various interactions within the network, and the following exposition is meant only as an illustration of certain interpretative potentialities of such a theory.

Let us start with considering an immunogenic substance which presents epitope E to the immune system. This epitope is recognized (with various degrees of precision) by a set of different antibody combining sites, or paratopes. I call this set p_1 and note that these paratopes occur on antibody and receptor molecules together with certain idiotopes, so that the set p_1 of paratopes is associated with a set i_1 of idiotopes. The symbol $p_1 i_1(E)$ denotes the total set of recognizing antibody molecules and potentially responding lymphocytes with respect to epitope E (see fig. 1). Within the network of the immune system, each paratope of the set p_1 recognizes a set of idiotopes, and the entire set p_1 recognizes an even larger set of idiotopes. This set i_2 of idiotopes is, as it were, the « internal image » of epitope E because it is recognized by the same set p_1 that recognizes E. The set i_2 is associated with a set p_2 of paratopes occurring on the molecules and cell receptors of the set $p_2 i_2(E)$.

Furthermore, each idiotope of the set $p_1 i_1(E)$ is recognized by a set of paratopes, so that the entire set i_1 is recognized by an even larger set p_3 of paratopes which occur together with a set i_3 of idiotopes on antibodies and lymphocytes of the anti-idiotypic set $p_3 i_3(E)$. We come in this way to ever larger sets that recognize or are recognized by previously defined sets within the network. As a first gross approximation to a functional network

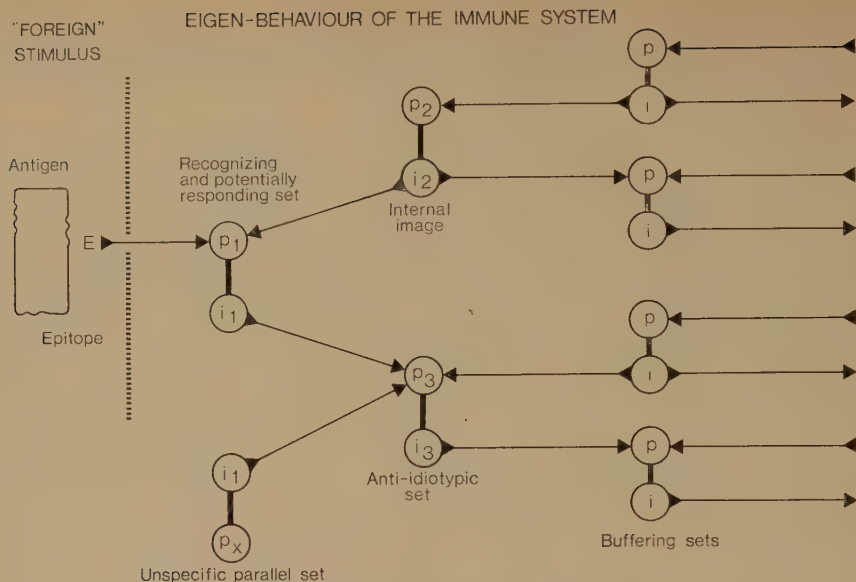


FIG. 1. — *Eigen-behaviour of the immune system.*

The immune system contains a set p_1 of combining sites (paratopes) on immunoglobulin molecules and on cell receptors which recognize a given epitope (E) of an antigen. This recognizing set includes the potentially responding lymphocytes. The molecules of set p_1 also display a set i_1 of idiotopes. Apart from recognizing the foreign epitope, the set p_1 likewise recognizes a set i_2 of idiotopes which thus constitutes, within the immune system, a kind of internal image of the foreign epitope. This set i_2 occurs in molecular association with a set p_2 of paratopes. Likewise, the set i_1 is recognized, within the immune system, by a set p_3 of paratopes which represent anti-idiotypic antibodies. Beside the recognizing set $p_1 i_1$ there is a parallel set $p_x i_1$ of immunoglobulins and cell receptors which display idiotopes of the set i_1 in molecular association with combining sites that do not fit the foreign epitope. As a first approximation, the arrows indicate a stimulatory effect when idiotopes are recognized by paratopes on cell receptors and a suppressive effect when paratopes recognize idiotopes on cell receptors. Successive groups of ever larger sets encompass the entire network of the immune system.

we could assume that the elements (molecules and cells) of the « internal image » set $p_2 i_2(E)$ are largely stimulatory towards the potentially responsive set $p_1 i_1(E)$ of lymphocytes, whereas the elements of the anti-idiotypic set $p_3 i_3(E)$ are largely inhibitory. In the eigen-behaviour of the immune system these opposing forces result in a balanced suppression which must be overcome in order to obtain an immune response to E.

I should now like to confront this first crude approach towards constructing a *network theory* with a number of phenomena and experimental findings, some of which were reported at this colloquium. The purpose is to show that a network theory might offer alternatives to current explanations.

1) *The immune response.* The elimination by the antigen of natural antibody of the set $p_1 i_1(E)$ removes its inhibitory effect on the « internal image » as well as its stimulatory effect on the anti-idiotypic set. Both effects favour an escape from suppression of cells of the set $p_1 i_1(E)$.

2) *Regulation*. The enhancement of $p_1i_1(E)$ will tend to reverse the above effects. The network attempts to restore its equilibrium.

3) *Low-zone tolerance*. Only a minimal fraction of natural antibody is removed, but in complex with antigen its idiotopes preferentially stimulate the suppressive anti-idiotypic set $p_3i_3(E)$.

4) *7S-inhibition*. The inhibition of the immune response by small amounts of passively given specific antibody may have to do with the recognition by this antibody of idiotopes of the internal image set $p_2i_2(E)$. The suppression of this stimulatory set may lead to non-responsiveness of $p_1i_1(E)$. This suggestion is reminiscent of the explanation offered by Rowley and Fitch [30] who have proposed that potentially antibody forming cells themselves carry an « image » of the antigen to which their antibody product is directed.

5) *Antigen-competition*. The modulations of the network required for the responses to two antigens may not be simultaneously possible.

6) *The Weigle phenomenon* [7]. Cells of the set $p_1i_1(E)$ remain suppressed when the inhibitory effect of $p_3i_3(E)$ continues to dominate over the stimulatory effect of $p_2i_2(E)$. The breakage of this tolerance could result if cross reacting antigen influences corresponding overlapping sets sufficiently.

7) *The Oudin-Cazenave enigma* [1], namely that antibodies produced against different non cross-reacting epitopes E_1 and E_2 of one antigenic molecule may have similar idiotypes. Considering the magnitudes of the paratope and idiope repertoires, this finding seems to make no sense within current theoretical concepts, since it would seem to be highly unlikely that two different, independently selected paratopes should occur on antibody molecules of similar idiope. In a network theory we might seek for an explanation along the following lines. The epitopes E_1 and E_2 are recognized by the two potentially responsive sets $p_1i_1(E_1)$ and $p_1i_1(E_2)$. Though non cross-reactivity implies that these two sets do not overlap with respect to p_1 , they may overlap with respect to i_1 . This possibility becomes more likely if we assume that a given paratope does not necessarily occur on all molecules in association with the same idiotopes, because then the set i_1 is larger than the set p_1 . Furthermore, the larger suppressive sets $p_3i_3(E_1)$ and $p_3i_3(E_2)$ may well overlap with respect to p_3 . If we now make the *ad hoc* assumption that such an overlap of paratopes, that these sets $p_3i_3(E_1)$ and $p_3i_3(E_2)$ have in common, will favour the escape from suppression of certain cells of the sets $p_1i_1(E_1)$ and $p_1i_1(E_2)$ which thereby become the major effectors of the antibody responses to both E_1 and E_2 , we have explained the Oudin-Cazenave findings without sacrificing the basic concept that an antigen-sensitive lymphocyte is precommitted and does not recognize an entire antigen molecule but only one epitope on this molecule.

8) *Unspecific response*. The production of unspecific immunoglobulin after an immunogenic stimulus has often been observed. Our chairman

this afternoon, Dr Humphrey, has devoted several studies to this problem [13]. How can we explain the report by Oudin [1] that such unspecific immunoglobulin molecules were found to be similar in idio type to the specific antibodies [5]? If we consider epitope E and the potentially responding set of lymphocytes that belong to the set $p_1 i_1(E)$ we may admit that certain idiotypes of this set i_1 occur also on molecules which have paratopes that do not belong to the set p_1 . Let us denote this parallel set by $p_x i_1(E)$. The release of lymphocytes of set $p_1 i_1(E)$ from suppression may at the same time release from suppression lymphocytes of the parallel set displaying the same idiotopes. These cells will then produce globulin molecules of this idio type but with combining sites of set p_x which are « unspecific » as they do not fit the immunizing epitope E.

The weakness of this incipient network theory lies in its lack of precision. The formal sets $p_1 i_1$, $p_2 i_2$, $p_3 i_3$, etc., refer to sets of antibody molecules that can occur both as freely circulating antibodies of all immunoglobulin classes and as functional receptors on T cells and B cells. This leaves an ambiguity in the answer to the question whether the relations between two sets is suppressive or stimulatory, or partly one and partly the other, and thus permits us to postulate interactions that suit our explanatory needs. The properties of all possible different types of interaction must therefore be clarified in order to be able to understand and eventually to manipulate the immune system. This appears to be the immense task facing immunology in the period ahead of us. Another question that needs consideration is the ontogenic development of the network in the adult immune system from a small germ-line determined, embryonic set of elements. We suspect that the initial shaping of the immune network is influenced to a large extent by the genes that cluster around the histocompatibility locus, and that this view will yield a more plausible explanation of the findings reported by Benacerraf and McDevitt [1] than postulating an entirely new second recognition system based on molecules that are not immunoglobulins. Partly because of polymorphisms in the cluster of genes at the histocompatibility locus, and partly because of a certain randomness in the ontogenic emergence of new paratopes and idiotopes, each individual will develop a different immune network. When established, the immune network must possess quite stable features. Early modulations of the network must be hard to undo later. This may be relevant to Fazekas de St Groth's concept of the original antigenic sin [9] and it suggests that immunological memory may be more firmly deposited in a persistent network modulation than in the persistence of a population of so-called « memory cells ».

I cannot pursue these matters further within the frame of my concluding remarks to this colloquium, except for saying that the clonal selection theory has regarded antigen-sensitive lymphocytes as independent cells among which selection operates in a way that is similar to the selection of

particular bacterial clones in a population of bacteria. Our proposal incorporates the clonal selection theory into the wider frame of a network, without sacrificing it.

VII. — THE IMMUNE SYSTEM AND THE NERVOUS SYSTEM

In finishing, I should only like to point out that the immune system, when viewed as a functional network dominated by a mainly suppressive Eigen-behaviour, but open to stimuli from the outside, bears a striking resemblance to the nervous system. These two systems stand out among all other organs of our body by their ability to respond adequately to an enormous variety of signals. Both systems display dichotomies and dualisms. The cells of both systems can receive as well as transmit signals. In both systems the signals can be either excitatory or inhibitory. The two systems penetrate most other tissues of our body, but they seem to be kept separate from each other by the so-called blood-brain barrier. The nervous system is a network of neurons in which the axon and the dendrites of one nerve cell form synaptic connections with sets of other nerve cells. In the human body there are about 10^{12} lymphocytes as compared to 10^{10} nerve cells. Lymphocytes are thus a hundred times more numerous than nerve cells. They do not need connections by fibres in order to form a network. As lymphocytes can move about freely, they can interact either by direct encounters or through the antibody molecules they release. The network resides in the ability of these elements to recognize as well as to be recognized. Like for the nervous system, the modulation of the network by foreign signals represents its adaptation to the outside world. Early imprints leave the deepest traces. Both systems thereby learn from experience and build up a memory that is sustained by reinforcement and that is deposited in persistent network modifications, which cannot be transmitted to our offspring.

These striking phenotypic analogies between the immune system and the nervous system may result from similarities in the sets of genes that govern their expression and regulation.

KEY-WORDS: Immun system Immunoglobulin, Lymphocyte. Genetics, Immunogenesis; Network, Clonal selection, Eigen-behaviour, idiotype.

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Finally, I am sure to speak for all participants in thanking the organizers of this colloquium, and particularly Professor Jacques Oudin, for having brought us together at this memorable occasion.

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